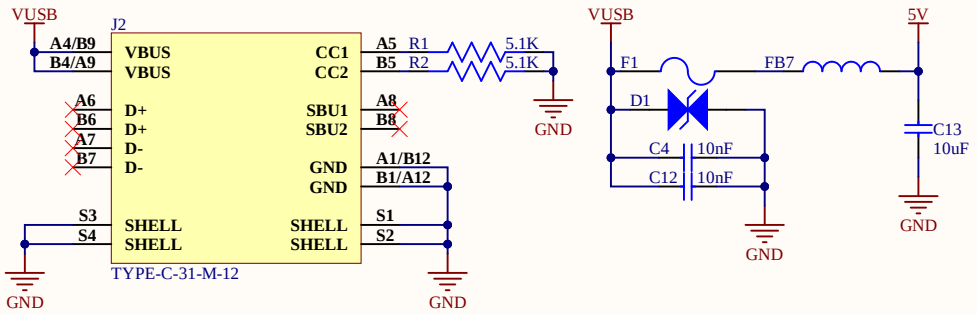


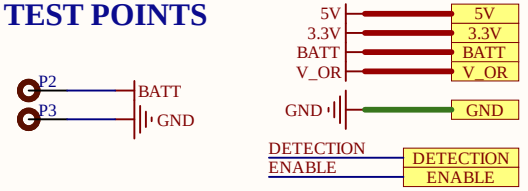
eSpark Consulting Group

Title: Top-Sheet.SchDoc		
File: IoT-Speaker.PrjPcb		
Date: 09/05/2023	Page	Num: 1
Drawn By: Huzaifa Ali	Page	Total: 5

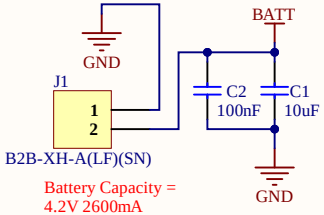
VUSB 5V CHARGE INPUT



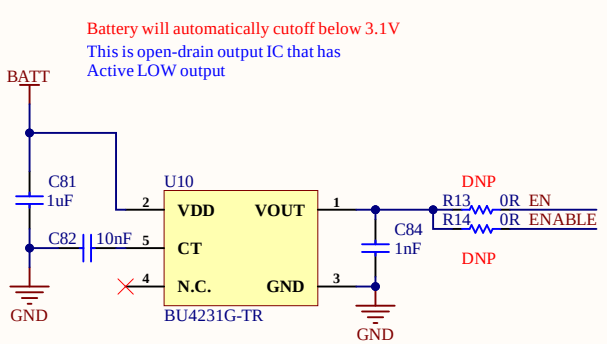
TEST POINTS



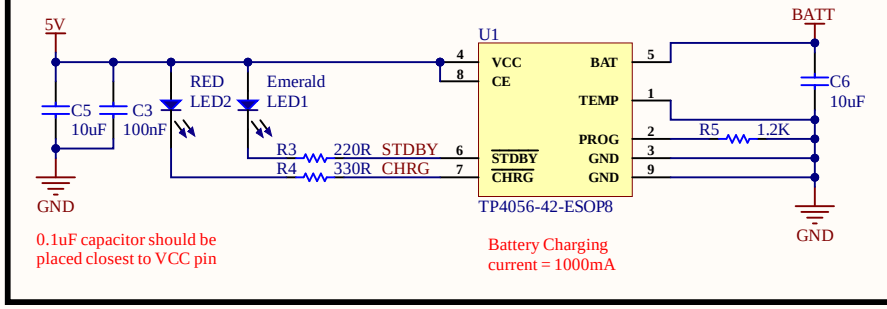
BATTERY INPUT



BATTERY SUPERVISOR CIRCUIT

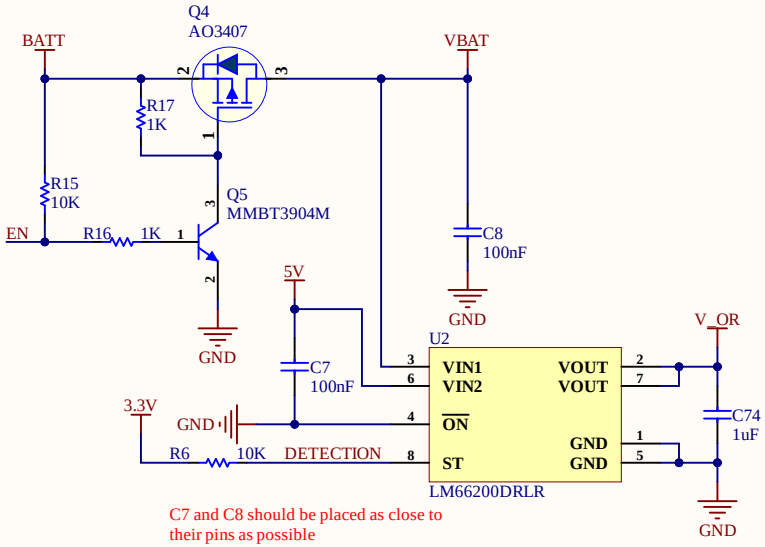


BATTERY CHARGING CIRCUIT



POWER SUPPLY ORING CIRCUITRY

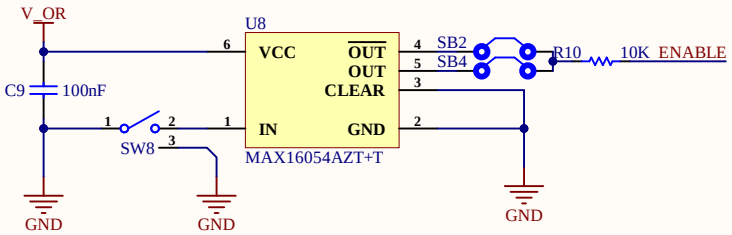
The BATT-VBAT circuit is only for testing and maybe removed in production. This is to evaluate if the battery supervisory doesn't quickly activate when charger is connected



eSpark Consulting Group

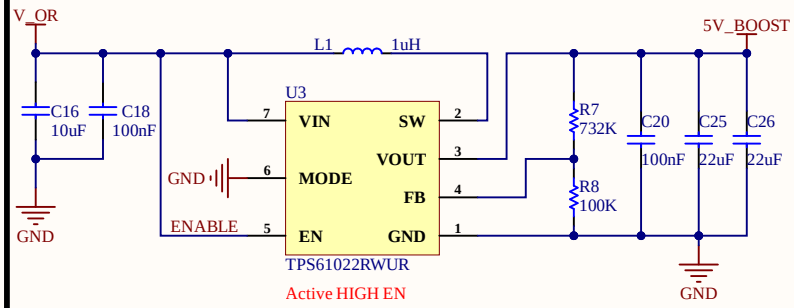
Title: Power-Sheet-1.SchDoc		
File: IoT-Speaker.PrjPcb		
Date: 09/05/2023	Page	Num: 2
Drawn By: Huzaifa Ali	Page	Total: 5

LATCH ON/OFF CONTROLLER



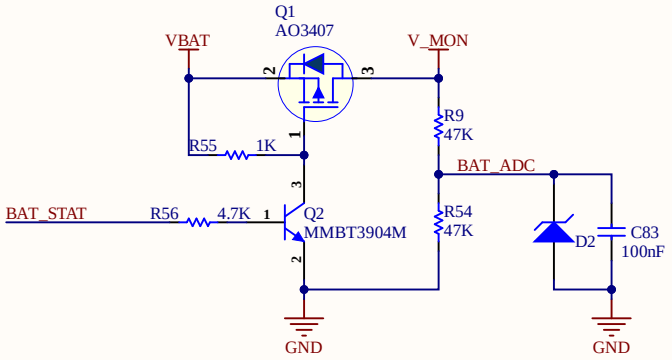
The OUT pin changes state whenever
LOW logic is received at IN pin

BOOST CONVERTER

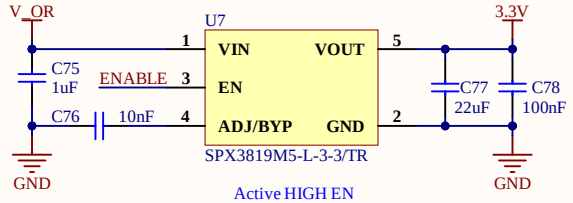


Active HIGH EN

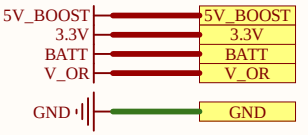
READ BATTERY VOLTAGE

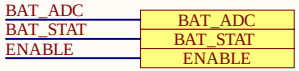


3V3 LDO

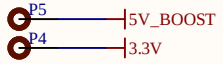


Active HIGH EN
3.3V trace current 500mA max.



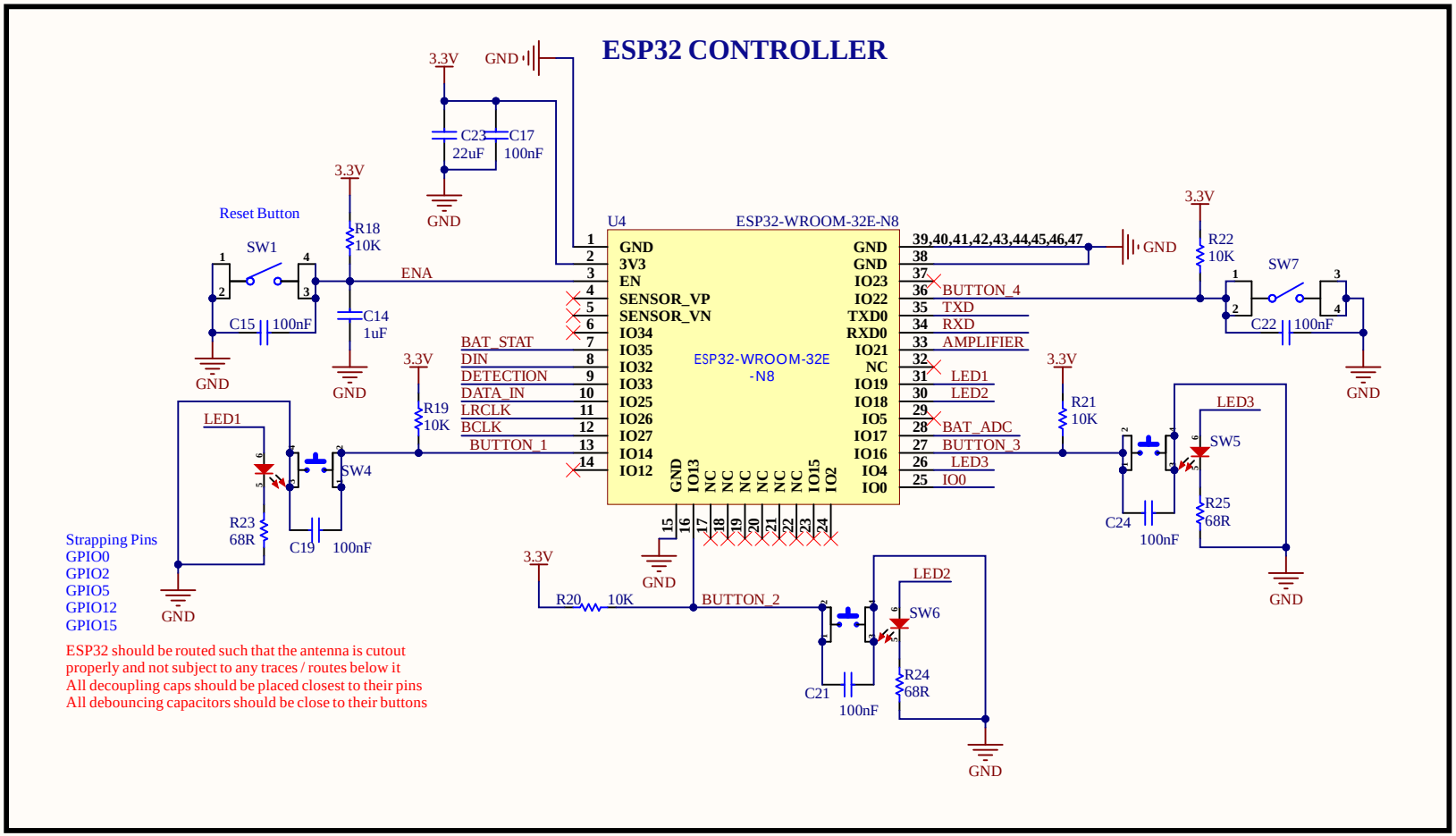


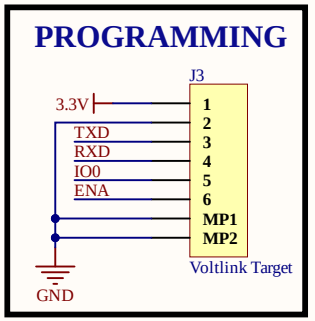
TEST POINTS

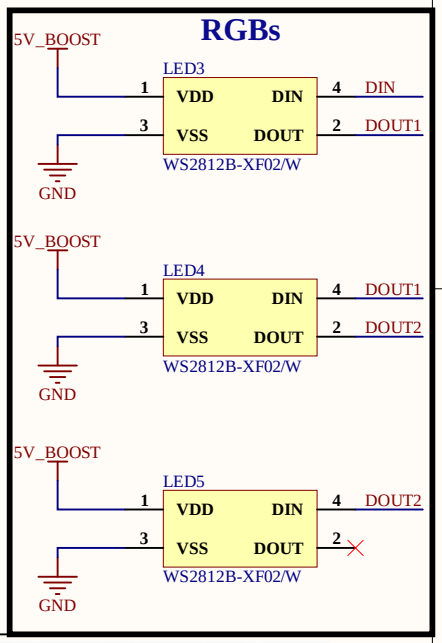


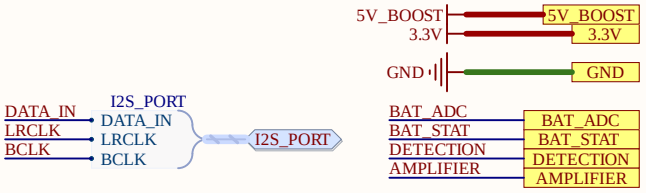
eSpark Consulting Group

Title: Power-Sheet-2.SchDoc	
File: IoT-Speaker.PrjPcb	
Date: 09/05/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 5



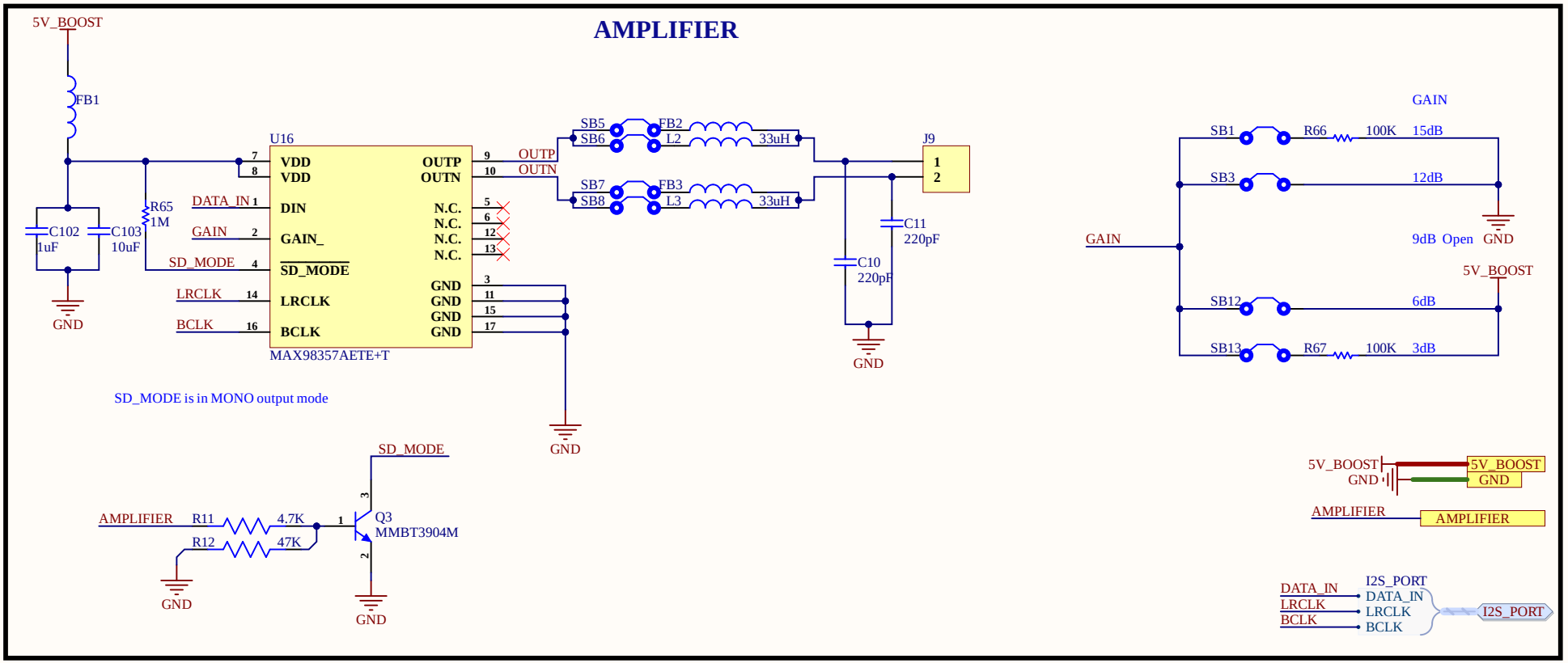






eSpark Consulting Group

Title: Controller-Sheet.SchDoc	
File: IoT-Speaker.PrjPcb	
Date: 09/05/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 5



eSpark Consulting Group

Title: Audio-Amplifier.SchDoc		
File: IoT-Speaker.PrjPcb		
Date: 09/05/2023	Page	Num: 5
Drawn By: Huzaifa Ali	Page	Total: 5